Questions-Gen — 系统架构 · 三阶段训练数据流

Training data → SFT → GRPO → DeepSeek distillation → validation → Hugging Face / Ollama

主数据流　　虚线框 = 架构分区

回退 / 反馈　　盒内 = 模块 + 职责

TRAINING DATA · problem mixture

Historical Problems
50% source distribution

Variants
30% conditional variation

Novel Problems
20% innovation target

Stage 1 SFT · QuestionsGenTrainer.py

Qwen3-14B
supervised fine-tuning

Stage-1 Checkpoint
stage1-fp-merged

Stage 2 GRPO · reward optimization

RewardCalculator
0.4 difficulty · 0.3 novelty
0.2 rigor · 0.1 diversity

NoveltyConstraint
variation / duplication guard

GRPO Trainer
group-relative optimization

Stage 3 Distill · teacher supervision

quality feedback

DeepSeekTeacher
DeepSeek-R1 reasoning teacher

Final Model
final-merged-16bit

VALIDATION

ModelValidator
single checkpoint

BatchValidator
algebra · geometry · calculus

QualityEvaluator
clarity · difficulty · originality

DEPLOYMENT

Hugging Face
three published stages

Ollama
local inference

questions-gen CLI
validate · batch · quality

主链：混合题库 → SFT → GRPO reward/novelty → DeepSeek-R1 distill → validate → publish。